

Lecture 12 – Time complexity

NTIN071 Automata and Grammars

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The translation, some modifications, and all errors are mine.*

Recap of Lecture 11

- reductions between decision problems
- the Halting problem is undecidable
- Rice's theorem: nontrivial semantic properties of programs are undecidable
- Undecidable problems about context-free grammars
- Source of undecidability: Post Correspondence Problem

CHAPTER 5: INTRO TO COMPLEXITY

Asymptotic notation

Big-O notation: Let $f, g : \mathbb{N} \rightarrow \mathbb{R}^+$. We say that $f(n) \in O(g(n))$, if there exist $C, n_0 \in \mathbb{N}^+$ such that

$$(\forall n \geq n_0) f(n) \leq C \cdot g(n)$$

i.e. $\limsup_{n \rightarrow \infty} \frac{f(n)}{g(n)} < \infty$. In that case we say that $g(n)$ is an [asymptotic] **upper bound** [up to a constant multiple] for $f(n)$.

Note: Often the imprecise term ‘upper bound’ is used; sometimes you will encounter $f(n) = O(g(n))$.

For example, $f(5n^3 + 2n^2 + 22n + 6) \in O(n^3)$ with $n_0 = 10, C = 6$.

Little-o notation: $f(n) \in o(g(n))$, if for all $c > 0$ there exists $n_0 \in \mathbb{N}^+$ so that $(\forall n \geq n_0) f(n) < c \cdot g(n)$, i.e. $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$. Then we say $f(n)$ is [asymptotically] **dominated** by $g(n)$.

Analogously for $\geq$ instead of $\leq$: $\Omega(g(n)), \omega(g(n))$

5.1 Time complexity

Classes of time complexity

Definition

Let M be a Turing machine that halts on every input. The **time complexity** of M is the function $f : \mathbb{N} \rightarrow \mathbb{N}$, where $f(n)$ is the maximum number of computation steps for inputs of length n .

Definition

For $t : \mathbb{N} \rightarrow \mathbb{R}^+$, **TIME**($t(n)$) is the class of all languages decidable by a TM of time complexity in $O(t(n))$ (i.e., always halts and for $|w| = n$ correctly answers in at most $O(t(n))$ steps).

NB: Here we mean the standard, single-tape, deterministic TM.

Example

Example ($L = \{0^i 1^j \mid i \geq 0\}$ is in $\text{TIME}(n^2)$)

1. check if the input is $0^i 1^j$, if a 0 follows a 1, reject (time $O(n)$)
2. return to the beginning: hidden in the constant $O(2n) = O(n)$
3. go through the 0s, in time $O(n^2)$
 - 3.1 rewrite the next 0 to X
 - 3.2 find the first 1, rewrite to X
 - 3.3 return to the beginning
4. if no more 0s, check that no more 1s remain and accept (if 1 found, reject) (time $O(n)$)

⋮

Can we do it faster?

Can we do it faster?

Idea: “compare the binary representations of i and j ”, $\log n$ bits, for each bit need to traverse through the word

Example ($L = \{0^i 1^j \mid i \geq 0\}$ is also in $\text{TIME}(n \log n)$)

1. check if the input is $0^i 1^j$ and even length (time $O(n)$)
2. iterate while there are 0s, in time $O(n \log n)$
 - 2.1 rewrite every other 0 to X , then every other 1 to X
 - 2.2 check if the number of remaining 0s+1s is even, if not, reject
3. if no more 0s, check that no more 1s and accept (time $O(n)$)

⋮

Can we do it even faster?

Time complexity and regular languages

Can we do it even faster? Not really.

Theorem

Every language decidable in time $o(n \log n)$ [on a single-tape, deterministic TM] is regular.

[We omit the proof. (It uses Myhill-Nerode theorem similarly to the proof that 2-way DFA only recognize regular languages.)]

Multi-tape TM

Example (Multi-tape TM for $L = \{0^i 1^i \mid i \geq 0\}$)

- copy 0s to Tape 2
- at first 1, switch state; erase 1 from Tape 1 & 0 from Tape 2
- accept if both tapes are erased

Lemma

Every multi-tape Turing Machine with time complexity $t(n)$ is equivalent to a [single-tape] Turing Machine with time complexity $O(t^2(n))$.

Proof: Simulation of n steps of a k -tape TM can be done in $O(n^2)$ moves since one step takes $4n + 2k$ moves (heads at most $2n$ fields apart, read, write, move head marks). □

Nondeterministic time complexity

The **time complexity** of a **nondeterministic** Turing machine that always halts is defined analogously: $f(n)$ is the maximum number of steps in **any branch** of the computation tree.

Definition

For $t : \mathbb{N} \rightarrow \mathbb{R}^+$, $\text{NTIME}(t(n))$ is the class of all languages decidable by a nondeterm. TM of time complexity in $O(t(n))$.

(An NTM **decides** L if halts on all inputs and recognizes L .)

Theorem

Any nondeterministic TM of time complexity $t(n) \geq n$, has a deterministic equivalent of time complexity in $2^{O(t(n))}$.

Corollary

If $t(n) \geq n$, then $\text{NTIME}(t(n)) \subseteq \text{TIME}(2^{O(t(n))})$.

Recall the construction: BFS of the computation graph, keep a queue of configurations to process.

- At most d possible transitions for any $(q, X) \in (Q \setminus F) \times \Gamma$.
- So after k steps at most d^k configurations.
- Processing one configuration can be 'hidden' in the constant.
- Therefore the simulation is in time:

$$O(t(n)d^{t(n)}) = 2^{O(t(n))}$$

- We need to simulate multiple tapes, but:

$$(2^{O(t(n))})^2 = 2^{O(2t(n))} = 2^{O(t(n))}$$



P vs. NP

The class P

Definition

Let **P** (also **PTIME**) be the class of all languages decidable in **polynomial time** by a [single-tape, deterministic] Turing machine:

$$P = \bigcup_k \text{TIME}(n^k)$$

- Path in a graph
- Primality of an integer (Agrawal, Kayal, Saxena 2002)
- Linear programming
- Horn-SAT

(The last two are P-complete under LOGSPACE reductions.)

Theorem ($CFL \subseteq P$)

Every context free language belongs to P.

Proof: Take a ChNF grammar for L . Given input ω , run the CYK algorithm (polynomial, in $O(n^3)$). □

The class NP: verifier-based definition

Definition

A **verifier** for a language L is an algorithm V such that:

$$L = \{w \mid \text{there exists a finite string } c \text{ such that } V \text{ accepts } \langle w, c \rangle\}$$

Such a c is called a **certificate**. It can be over any alphabet!

Complexity of verifiers is only considered wrt. the length of w : a **polynomial verifier** must halt in time $O(|w|^k)$ for some $k > 0$.

Then we can assume the certificate has polynomial length (otherwise the verifier cannot even read it).

Definition

NP is the class of all languages that have a polynomial verifier.

That is, there is an algorithm that works in time polynomial in $|w|$ and when given $w \in L$ and a certificate c validates that c is a valid certificate for $w \in L$.

Hamiltonian path

A **Hamiltonian path** in a directed graph G is a directed path P that visits each vertex of G exactly once.

$$\text{HAMPATH} = \{\langle G \rangle \mid G \text{ contains a Hamiltonian path}\}$$

- complexity for graphs can be measured just wrt. $|V|$ ($|E|$ is at most quadratic, thus polynomial)
- the **certificate** is the path (sequence of vertices)
- the algorithm verifies that the sequence is indeed a path containing each vertex exactly once; this can be easily done in polynomial time wrt. $|V|$
- for $\overline{\text{HAMPATH}}$ we do not know whether a polynomial verifier exists (we only know the problem is in EXPTIME)

The class NP: nondeterminism-based definition

Definition

NP is the class of all languages that have a polynomial verifier.

Theorem

$$\text{NP} = \bigcup_k \text{NTIME}(n^k).$$

Idea: convert a verifier to a nondeterministic TM, and vice versa

⇒ the NTM guesses the certificate, then simulates the verifier

⇐ the verifier takes as a certificate the accepting path of the NTM (more precisely, the sequence of nondeterministic choices that leads to acceptance), then simulates the NTM

Proof

$\text{NP} \subseteq \bigcup_k \text{NTIME}(n^k)$: Let $L \in \text{NP}$ and take a polynomial verifier V for L , say it works in time $C \cdot |\omega|^k$. Construct an NTM M :

Given input ω :

- nondeterministically guess a certificate c (of $|c| \leq C \cdot |\omega|^k$)
- simulate V on input $\langle \omega, c \rangle$
- if V accepted, M accepts

$\bigcup_k \text{NTIME}(n^k) \subseteq \text{NP}$: Let $L \in \text{NTIME}(n^k)$, i.e., $L = L(M)$ for an NTM M working in time $O(n^k)$. Construct a polynomial verifier V :

Given input $\langle w, c \rangle$, interpret c as sequence of choices: $c_i = j$ means “at step i use j th possible transition” (order as in $\text{code}(M)$)

- simulate M on input w
- at each step i choose the c_i th possible transition
- accept if this computation path leads to acceptance



Example: CLIQUE is in NP

$\text{CLIQUE} = \{\langle G, k \rangle \mid G \text{ is a graph which contains } K_k \text{ as a subgraph}\}$

Polynomial verifier for CLIQUE: input $\langle \langle G, k \rangle, c \rangle$

- interpret the certificate c as a list of vertices
- check that c contains k vertices
- check that c induces a complete subgraph of G

Nondeterministic TM deciding CLIQUE: input $\langle G, k \rangle$

- nondeterministically choose a k -element subset $c \subseteq V$
- check that c induces a complete subgraph of G

Polynomial-time reductions and NP-completeness

Polynomial-time reducibility

Recall the notion of **reduction** between decision problems. Now we additionally require that the algorithm is polynomial-time:

A [total] function $f : \Sigma^* \rightarrow \Delta^*$ is **polynomial-time computable**, if there exists a [deterministic] Turing Machine M and $C, k > 0$ such that for each $\omega \in \Sigma^*$, M halts in at most $C \cdot |\omega|^k$ steps with $f(\omega) \in \Delta^*$ being the non-blank contents of its tape.

Definition

A language $A \subseteq \Sigma^*$ is **polynomial-time reducible** to a language $B \subseteq \Delta^*$, $A \leq_P B$, if there exists a polynomial-time computable function $f : \Sigma^* \rightarrow \Delta^*$ such that for all $\omega \in \Sigma^*$:

$$\omega \in A \Leftrightarrow f(\omega) \in B$$

Then we call f a **polynomial-time reduction** from A to B .

Example: Hamiltonian path from source to target

- $\text{HAMPATH} = \{\langle G \rangle \mid G \text{ contains a Hamiltonian path}\}$
- $\text{st-HAMPATH} = \{\langle G, s, t \rangle \mid G \text{ has a H. path from } s \text{ to } t\}$

Example

HAMPATH and st-HAMPATH are **polynomial-time interreducible**, i.e. each polynomial-time reduces to the other.

The reduction $\text{HAMPATH} \leq_P \text{st-HAMPATH}$:

Given G create G' by adding new vertices s, t and all edges from s to V_G and from V_G to t ; define $f(\langle G \rangle) = \langle G', s, t \rangle$

$$\langle G \rangle \in \text{HAMPATH} \Leftrightarrow \langle G', s, t \rangle \in \text{st-HAMPATH}$$

The reduction $\text{st-HAMPATH} \leq_P \text{HAMPATH}$: construct G' by adding new vertices s', t' , edges $s' \rightarrow s, t \rightarrow t'$; $f(\langle G, s, t \rangle) = \langle G' \rangle$

Example: 3SAT is polynomial-time reducible to CLIQUE

A propositional formula is in **CNF** if it is a conjunction of clauses, and **3-CNF** if each clause contains exactly 3 literals.

- **SAT** = $\{\langle \varphi \rangle \mid \varphi \text{ is a satisfiable CNF formula}\}$
- **3SAT** = $\{\langle \varphi \rangle \mid \varphi \text{ is a satisfiable 3-CNF formula}\}$

Theorem

3SAT is polynomial-time reducible to CLIQUE.

Proof: Vertices are occurrences of literals (three vertices per clause). Include all edges except for:

- between vertices from the same clause
- between a variable and its negation (x and $\neg x$)

Set $k = \# \text{clauses}$. Note: Exactly one literal per clause selected. $\square$

Exercise: 3SAT is polynomial-time interreducible with SAT.

NP-completeness

Definition

A language B is **NP-complete**, if $B \in \text{NP}$ and every language $A \in \text{NP}$ is polynomial-time reducible to B .

Observe:

- If some NP-complete B is in P , then $P = \text{NP}$.
- If B is NP-complete, $B \leq_P C$ and $C \in \text{NP}$, then C is NP-complete. (Why? $\leq_P$ is transitive.)

Exercise: Prove that $P \neq \text{NP}$.

Note: We call B **NP-hard** if some/every NP-complete problem reduces to it (but B is not necessarily in NP). This makes sense even for problems that are not 'decision' problems (for example 'counting problems').

Cook-Levin theorem

Theorem (Cook, Karp, Levin ca. 1971)

SAT is NP-complete.

Proof: **SAT is in NP:** nondeterministic TM that guesses a satisfying assignment, verifies it satisfies φ (in polynomial time).

SAT is NP-hard: The idea is to encode computation of a (nondeterministic) TM as a SAT instance.

Take any $L \in NP$ and let M be an NTM that decides L in time $n^k - 3$ for some k (for simplicity assume one-way infinite tape).

Describe the computation of M on input w in a table [next slide].

Proof cont'd: computation in a table

Construct a $n^k \times n^k$ table where each row corresponds to a configuration for computation of M on input w .

The first row describes the initial config, each next row obtained by one move, or by zero moves (if we already halted).

We bookend the configurations by $\#$'s.

#	q_0	w_1	w_2	$\dots$	w_n	-	-	$\dots$	#
#									#
#	$\vdots$								#
#	$\vdots$								#
#									#

We inspect the table by 2×3 frames:

[next slide]

Proof cont'd: legal frames

We describe **legal frames**, here only a selection ($a, b, c, d \in \Gamma$):

$$\delta(q_1, b) \ni (q_2, c, L)$$

a	q_1	b
q_2	a	c

$$\delta(q_1, b) \ni (q_2, c, R)$$

a	q_1	b
a	c	q_2

$$\delta(q_1, b) \ni (q_2, c, R)$$

d	a	q_1
d	a	c

$$\delta(-, -) \ni (q_2, -, L)$$

a	b	c
a	b	q_2

$$\delta(-, -) \ni (-, c, L)$$

a	b	d
c	b	d

$$\text{no change}$$

#	a	b
#	a	b

Claim: If 1st row contains initial config and each frame legal, then each row corresponds to a valid move (or is a copy once we halted).

- each row has at most one state
- legal frames with a state correspond to valid moves
- if no state, transfer without change
- proof technical, we omit some of the details

Proof cont'd: describe table by CNF formula

For each (i,j) and $a \in \Gamma \cup Q \cup \{\#\}$ create a boolean variable $x_{i,j,a}$.

$$\varphi = \varphi_{\text{cell}} \wedge \varphi_{\text{start}} \wedge \varphi_{\text{move}} \wedge \varphi_{\text{accept}}$$

Each cell has exactly one symbol:

$$\varphi_{\text{cell}} = \bigwedge_{1 \leq i,j \leq n^k} \left(\left(\bigvee_{a \in \Gamma \cup Q \cup \{\#\}} x_{i,j,a} \right) \wedge \bigwedge_{s \neq t \in \Gamma \cup Q \cup \{\#\}} (\overline{x_{i,j,s}} \vee \overline{x_{i,j,t}}) \right)$$

The first row is the initial configuration:

$$\varphi_{\text{start}} = x_{1,1,\#} \wedge x_{1,2,q_0} \wedge x_{1,3,w_1} \wedge \dots \wedge x_{1,n+2,w_n} \wedge x_{1,n+3,-} \wedge \dots \wedge x_{1,n^k,\#}$$

We got to an accepting state:

$$\varphi_{\text{accept}} = \bigvee_{1 \leq i,j \leq n^k} x_{i,j,q_F}$$

Proof finished

Legality of the table is a conjunction of legality of all frames:

$$\varphi_{\text{move}} = \bigwedge_{1 \leq i < n^k, 1 \leq j < n^k} \varphi_{i,j}$$

Legality of one frame is a disjunction over all legal frames:

$$\varphi_{i,j} = \bigvee_{\substack{(a_1, \dots, a_6) \\ \in \text{LEGAL}}} (x_{i,j-1,a_1} \wedge x_{i,j,a_2} \wedge x_{i,j+1,a_3} \wedge x_{i+1,j-1,a_4} \wedge x_{i+1,j,a_5} \wedge x_{i+1,j+1,a_6})$$

This is not in CNF but can be converted: $\varphi_{i,j}$ doesn't depend on the input, only on the TM, except need to write indices i, j ($\log n$)

Claim: reduction has polynomial time complexity, $O(n^{2k} \log n)$.

- we need $\log n$ to write indices of variables
- φ_{cell} in $O(n^{2k} \log n)$, go through all cells
- φ_{start} in $O(n^k \log n)$, go through the first row
- $\varphi_{\text{move}}, \varphi_{\text{accept}}$ in $O(n^{2k} \log n)$, all cells

The class co-NP, tautology

Definition

A language $L \subseteq \Sigma^*$ belongs to the class **co-NP**, if and only if its complement $\bar{L} = \Sigma^* - L$ belongs to NP.

NP = certificate of success, co-NP = certificate of failure

Example

TAUT is the decision problem whether a given propositional formula is a tautology (i.e., satisfied by every assignment).

Theorem

TAUT *is* co-NP-complete.

Proof: observe $\overline{\text{TAUT}} \in \text{NP}$ (guess falsifying assignment); note that $A \leq_P B$ iff $\bar{A} \leq_P \bar{B}$; so TAUT is co-NP-hard iff $\overline{\text{TAUT}}$ is NP-hard. But $\text{SAT} \leq_P \overline{\text{TAUT}}$ (φ satisf. iff $\neg\varphi$ not a tautology) $\square$

- it is expected that $\text{NP} \neq \text{co-NP}$ (implies $P \neq \text{NP}$ but not iff)
- P is contained in $\text{NP} \cap \text{co-NP}$
- it is **unknown** whether $P = \text{NP} \cap \text{co-NP}$

Example (Integer factorization: decision version)

$\text{FACTOR} = \{ \langle n, k \rangle \mid n \text{ has a prime divisor } \leq k \}$

- **in NP:** guess p ; verify $p \leq k$ and $p \mid n$ in polynomial time
- **in co-NP:** guess the prime factorization $p_1^{e_1}, \dots, p_m^{e_m}$; verify that product equals n , and that each p_i is $> k$ and prime (AKS primality test is in P)

$\Rightarrow \text{FACTOR} \in \text{NP} \cap \text{co-NP}$

?? $\text{FACTOR} \in P$

Exponential time complexity

$$\text{EXPTIME} = \bigcup_k \text{TIME}(2^{n^k})$$

complete problems:

- Given $\langle M, w, k \rangle$, does DTM M halt on input w in $\leq k$ steps?
- Is a given generalized $(n \times n)$ chess position winning for white?

$$\text{NEXPTIME} = \bigcup_k \text{NTIME}(2^{n^k})$$

Observe: $P \subseteq NP \subseteq \text{EXPTIME} \subseteq \text{NEXPTIME}$

The **Time Hierarchy Theorem** (“given more time, TMs can solve more problems”) implies $P \subsetneq \text{EXPTIME}$ and $NP \subsetneq \text{NEXPTIME}$.

Summary of Lecture 12

- time complexity, for TM as well as NTM
- the class P
- the class NP: verifier-based and nondeterminism-based definitions
- polynomial-time reductions
- NP-complete problems
- Cook-Levin theorem: SAT is NP-complete
- the class co-NP
- exponential time complexity